

Status-Coherent Clique Mining in Signed Networks

Layla Payton, TaiNing Wang
Loyola University Chicago
tpayton1@luc.edu, twang12@luc.edu

Problem

Signed graphs encode **positive and negative relationships** between entities and arise naturally in social network systems, trust systems, and opinion platforms. Existing work on clique mining in signed graphs relies on structural **balance theory** as the coherence criterion, but balance theory is known to be a poor fit for directed networks. **Status theory**, which interprets a positive edge from A to B as A asserting higher standing for B, better explains empirically observed link patterns in **directed** signed graphs.

We study the problem of mining **maximum cliques** that are coherent under status theory rather than structural balance. This shift introduces concrete systems challenges: status coherence requires verifying consistent directed status orderings within **candidate cliques**, which demands efficient triangle counting, directed triad enumeration, and constraint propagation across partial solutions. Standard pruning and bounding techniques developed for unsigned or balance-based clique search do not transfer cleanly, as the status coherence constraint lacks the monotone decomposability that makes those approaches effective.

Balance & Status Theory

Structural Balance Theory: Social dynamics can be used as a means to determine balanced and unbalanced triads in signed networks. The balanced combinations of edge signs in triads are (+++): a friend of a friend is a friend, and (+--): an enemy of an enemy is a friend. Unbalanced combinations go against the social dynamics of status theory, they are (++-) and (---). An extension of Balance Theory states that a signed subgraph is structural balanced if its vertices can be partitioned into two sets such that all edges inside each partition have positive signs and all cross-partition edges have negative signs.

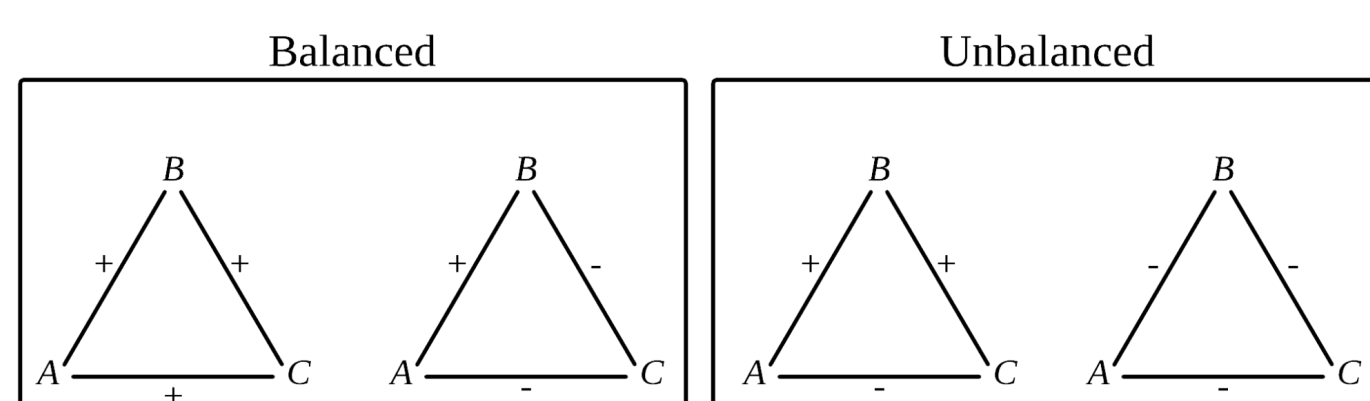


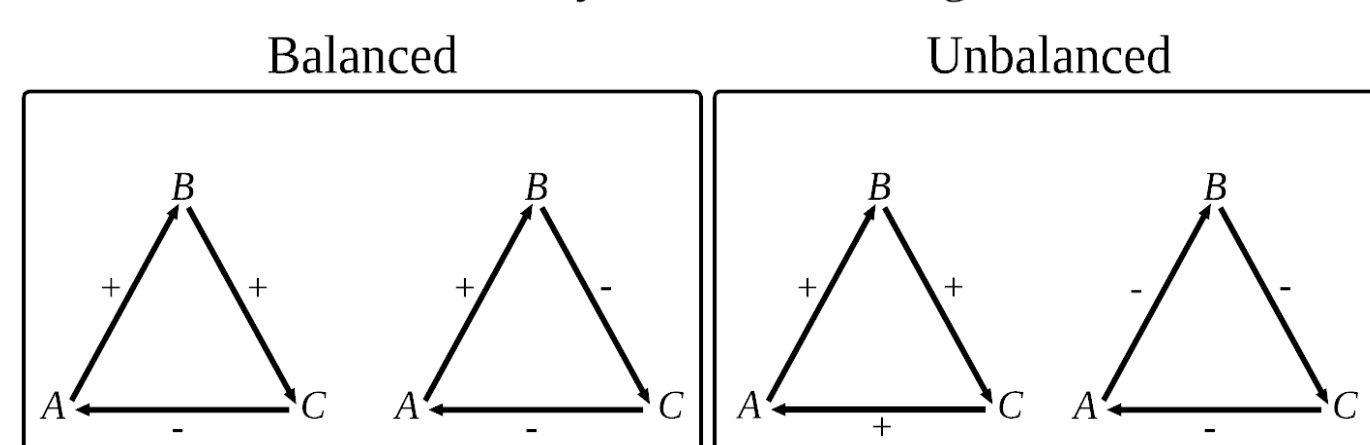
Figure 1: Structural Balance Theory Triangles

Status Theory: A positive directed edge indicates that the creator of the edge views the recipient as having higher status; and a negative directed edge indicates that the recipient is viewed as having lower status.

Status-based triads may follow two possible orientations:

- **Round Cycle:** All nodes have one in-edge and one out-edge.
- **Fan Cycle:** One node has two out-edges, one has two in-edges, and one has one in-edge and one out-edge.

Round-Cycle Status Triangles



Fan-Cycle Status Triangles

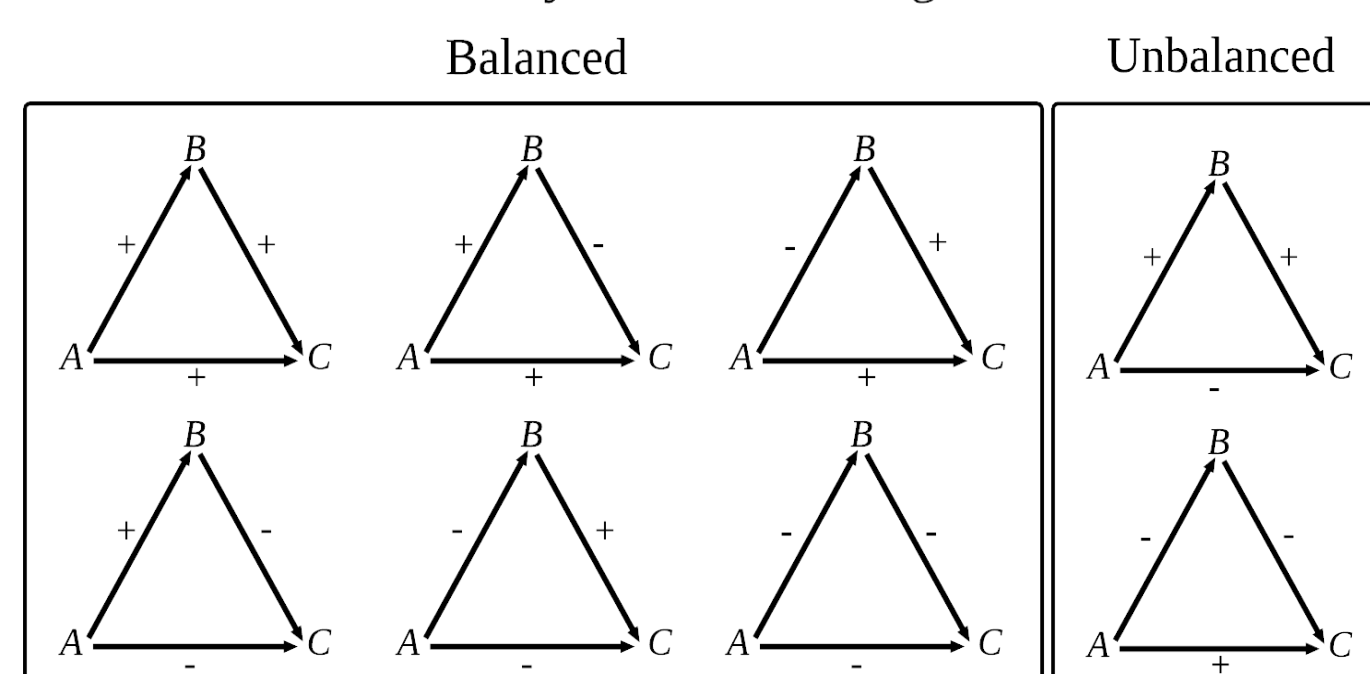


Figure 2: Balanced vs. Unbalanced Status-based triads

Finding Maximum Clique

Balanced Clique: A clique is balanced if its set of vertices C can be partitioned into C_L and C_R such that all negative edges are between them.

Maximum Balanced Clique Computation: Find the largest balanced clique C^* in graph G in which C_L^* and C_R^* are greater than a size threshold.

To optimize and make the problem of Maximum Balanced Clique Computation possible, **edge** and **vertex reduction** methods are implemented, which prunes vertices based on **degree threshold** (requiring a set amount of edges to other nodes) and edges based on **triangle participation** (edge is part of enough of both balanced and unbalanced triangles).

Status Theory Incorporation

Rather than partitioning cliques according to negative edges, we propose splitting cliques by edges that are members of unbalanced status triangles within a directed graph. To achieve **Maximum Status Clique Computation**, we implement a set of status-coherent trimming criteria through status-based edge and vertex reduction.

We propose a set of **Theoretical Status Theory Guarantees** which allow us to predict whether a triangle is balanced based off two edges, if a third edge were to exist. This allows us to optimize by trimming vertices in the vertex reduction method by comparing to these guarantees.

Theorem 1: Theoretical Status Theory Guarantees

if ($sign(u \rightarrow v) = sign(u \rightarrow w)$) $\Rightarrow \triangle uvw$ is balanced

if ($sign(v \rightarrow u) = sign(w \rightarrow u)$) $\Rightarrow \triangle uvw$ is balanced

if ($sign(u \rightarrow v) \neq sign(w \rightarrow u)$) $\Rightarrow \triangle uvw$ is balanced

Status-based triangles are enumerated by counting according to 4 different groups: **Round-Balanced**, **Round-Unbalanced**, **Fan-Balanced**, **Fan-Unbalanced** by checking the following criteria:

Check round or fan cycle:

If each node has exactly one out edge: round-cycle. If one node has two out edges, and one has zero out edges: fan-cycle.

Check the balance of each:

round_balanced($\triangle uvw$) = $\neg (u \rightarrow v = v \rightarrow w) \wedge (v \rightarrow w = w \rightarrow u)$

fan_balanced($\triangle uvw$) = $\neg (u \rightarrow v = v \rightarrow w) \wedge (u \rightarrow v \neq u \rightarrow w)$

This triangle enumeration criteria allows us to perform edge reduction by removing edges which are not involved in a sufficient amount of status-based triangles. Implementing a status-based edge reduction caused the runtime to be reduced by around 60%.

Summary

The problem of maximum balanced clique computation in an undirected network uses the popular Structural Balance Theory. However, in a directed environment, Structural Balance Theory may not return good results. We propose the implementation of Status Theory in its place, creating a bipartite on status non-adherence rather than negative edges. We have developed status-based methods for edge and vertex reduction, leveraging a system of theoretical status-theory guarantees to remove unpromising candidate edges and vertices from the calculation.

As future work, we hope to further develop graph management and analysis techniques on signed networks by leveraging both Structural Balance Theory and Status Theory.